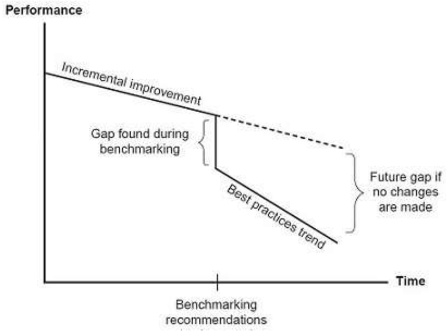
WHAT IS BENCHMARKING?

Quality Glossary Definition: Benchmarking

Benchmarking is defined as the process of measuring products, services, and processes against those of organizations known to be leaders in one or more aspects of their operations. Benchmarking provides necessary insights to help you understand how your organization compares with similar organizations, even if they are in a different business or have a different group of customers.

Benchmarking can also help organizations identify areas, systems, or processes for improvements—either incremental (continuous) improvements or dramatic (business process re-engineering) improvements.



ncremental Quality Improvement vs Benchmarking Breakthroughs

Benchmarking has been classified into two distinct categories: technical and competitive. The House of Quality matrix and Gantt charts are often used to plot the benchmarking evaluation.

- Technical benchmarking
- Competitive benchmarking
- Benchmarking procedure
- Benchmarking example
- Benchmarking studies
- Benchmarking resources

TECHNICAL BENCHMARKING

Technical benchmarking is performed by design staff to determine the capabilities of products or services, especially in comparison to the products or services of leading competitors. For example, on a scale of one to four, four being best, how do designers rank the properties of your organization’s products or services? If you cannot obtain hard data, the design efforts may be insufficient, and products or services may be inadequate to be competitive.

COMPETITIVE BENCHMARKING

Competitive benchmarking compares how well (or poorly) an organization is doing with respect to the leading competition, especially with respect to critically important attributes, functions, or values associated with the organization’s products or services. For example, on a scale of one to four, four being best, how do customers rank your organization’s products or services compared to those of the leading competition? If you cannot obtain hard data, marketing efforts may be misdirected and design efforts misguided.

BENCHMARKING PROCEDURE

Considerations

- Before an organization can achieve the full benefits of benchmarking, its own processes must be clearly understood and under control.
- Benchmarking studies require significant investments of manpower and time, so management must champion the process all the way through, including being ready and willing to make changes based on what is learned.
- Too broad a scope dooms the project to failure. A subject that is not critical to the organization's success won't return enough benefits to make the study worthwhile.
- Inadequate resources can also doom a benchmarking study by underestimating the effort involved or inadequate planning. The better you prepare, the more efficient your study will be.

Plan

1. Define a tightly focused subject of the benchmarking study. Choose an issue critical to the organization's success.
2. Form a cross-functional team. During Step 1 and 2, management's goals and support for the study must be firmly established.
3. Study your own process. Know how the work is done and measurements of the output.
4. Identify partner organizations that may have best practices.

Collect

5. Collect information directly from partner organizations. Collect both process descriptions and numeric data, using questionnaires, telephone interviews, and/or site visits.

Analyze

6. Compare the collected data, both numeric and descriptive.
7. Determine gaps between your performance measurements and those of your partners.
8. Determine the differences in practices that cause the gaps.

Adapt

9. Develop goals for your organization's process.
10. Develop action plans to achieve those goals.
11. Implement and monitor plans.

BENCHMARKING EXAMPLE

Carleton University's department of housing began a project to improve the process by which students apply for and are assigned housing. This project had three main goals:

1. Reduce residence vacancy rates to below 1%
2. Improve student satisfaction levels
3. Make the best possible use of employee time and effort

The department chose benchmarking as the basis for the improvement project after realizing that other universities already had better processes Carleton could learn from and quickly implement without needing to re-engineer the process from scratch.

The cross-functional team started by mapping the existing residence application process to establish a baseline, focusing attention on the process used by new first-year students (left side, Figure 1).

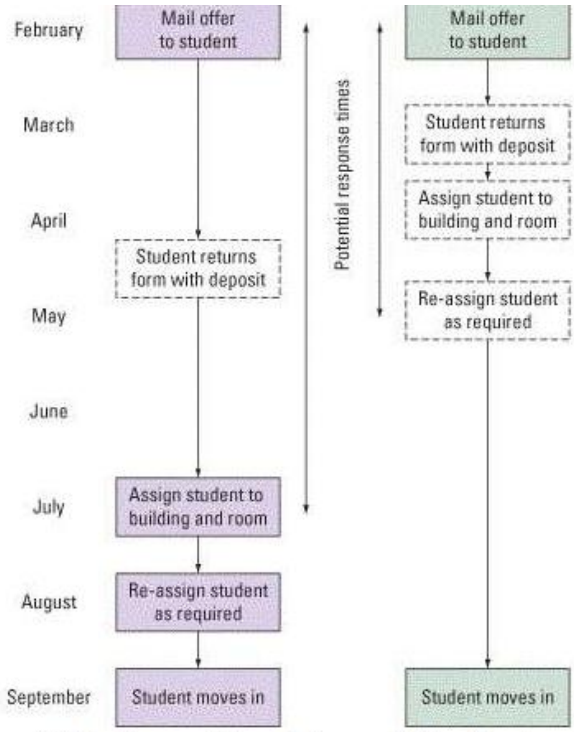


Figure 1: Main Process Steps Before and After Benchmarking

The project team then conducted focus groups with students living in residence, asking them to describe their expectations of the assignment service process, their perceptions of the services they actually received, and any gaps between the two.

The team also reviewed websites of housing offices at 24 international universities, followed by a telephone survey of 12 of these to find out more about their systems. The team then conducted site visits to four universities seen as leaders in dealing with residence applications to learn about their current processes and discuss the relative advantages and disadvantages of each one.

The team presented its benchmarking report to management, describing the best practices seen at other universities and making specific recommendations for adapting them for Carleton (right side, Figure 1). Carleton implemented many of the recommended changes and is seeing the benefits (Figure 2).

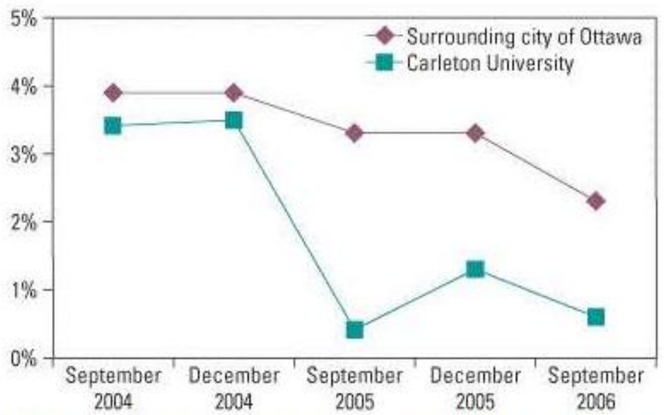


Figure 2: Benchmarking Average Vacancy Rates for Residences

Benchmarking is a strategic management approach that organisations use to gain a competitive edge by comparing their practices, processes, and performance metrics with those of their industry counterparts or top performers. It's a powerful tool that allows companies to identify areas for improvement, set performance targets, and implement effective strategies to enhance overall organisational performance. By analyzing data and information obtained from benchmarking partners, businesses can pinpoint performance gaps and discover best practices that can be adopted.

Steps of Benchmarking

Benchmarking involves a series of systematic steps that organisations can follow to effectively compare their performance and practices with industry leaders or competitors. The key steps involved in benchmarking are:

1. **Define the Focus:** Clearly identify the specific area or process that you intend to benchmark. Whether it's a particular function within your

organisation or a specific aspect of your industry, having a clear focus will ensure a targeted approach to improvement.

2. **Select Benchmarking Partners:** Identify organisations that excel in the chosen area and can serve as valuable benchmarks. Look for both direct competitors and companies from different industries known for their best practices. This diverse selection will provide a broader perspective and fresh insights for your improvement efforts.
3. **Gather Data and Information:** Collect relevant data and information from your benchmarking partners. Employ various methods such as surveys, interviews, site visits, or accessing publicly available reports. It's crucial to ensure the accuracy and comprehensiveness of the data, focusing specifically on the benchmarks you have identified.
4. **Analyze and Compare:** Analyze the collected data and compare it with your organisation's own performance. Identify gaps and differences in processes, practices, and performance metrics. This analysis will enable you to gain a deeper understanding of areas for improvement and learn from the best practices of your benchmarking partners.
5. **Set Performance Targets:** Based on the insights gained from the benchmarking analysis, establish specific performance targets and goals for your organisation. These targets should be challenging yet attainable, aligning closely with your strategic objectives. Clear targets provide a roadmap for your improvement efforts.
6. **Develop an Action Plan:** Create a comprehensive action plan that outlines the specific steps and initiatives required to bridge performance gaps and achieve the set targets. Ensure the action plan includes well-defined timelines, assigned responsibilities, required resources, and key milestones. Tailor the plan to suit the unique needs and capabilities of your organisation.
7. **Implement and Monitor:** Put your action plan into motion and execute the identified improvements within your organisation. Regularly monitor progress, tracking relevant performance indicators and metrics tied to your benchmarking focus. Ongoing monitoring allows for timely assessment of the effectiveness of implemented changes and enables necessary adjustments if required.
8. **Learn and Iterate:** Benchmarking is an iterative process that fosters continuous learning and improvement. Evaluate the outcomes of the implemented changes, draw insights from the results, and identify additional areas for enhancement. Leverage the knowledge gained through benchmarking to refine your processes, practices, and overall performance.

Types of Benchmarking

Benchmarking can be categorized into different types based on the nature of the comparison and the sources of benchmarking partners. Some common types of benchmarking are:

1. **Internal Benchmarking:** This type involves comparing performance and practices within different departments or units of the same organisation. It enables organisations to leverage best practices and successes from within their own ranks, promoting collaboration and knowledge sharing throughout the organisation.
2. **Competitive Benchmarking:** Competitive benchmarking entails comparing an organisation's performance and practices against direct competitors in the industry. The purpose is to identify areas where the organisation may be falling behind and to learn from the best practices of competitors. This type of benchmarking provides insights into industry standards, customer expectations, and competitive advantages.
3. **Functional Benchmarking:** Functional benchmarking focuses on specific functions or processes within an organisation. It involves comparing similar functions across different industries to discover innovative practices and process improvements. For instance, a manufacturing company might benchmark its supply chain management practices against a leading logistics company to enhance efficiency.
4. **Strategic Benchmarking:** Strategic benchmarking involves comparing long-term strategies, goals, and business models of organisations in similar or related industries. The objective is to learn from successful strategic approaches and identify opportunities for innovation and differentiation. Strategic benchmarking helps organisations align their strategic direction, make informed decisions, and adapt to evolving market dynamics.
5. **Performance Benchmarking:** Performance benchmarking concentrates on comparing key performance indicators (KPIs) and metrics across organisations in the same industry or sector. It enables organisations to understand their relative performance in terms of productivity, efficiency, quality, customer satisfaction, and financial performance. Performance benchmarking assists in setting realistic targets, tracking progress, and driving continuous improvement.
6. **International Benchmarking:** This type of benchmarking involves comparing performance and practices across national or international boundaries. Organisations can learn from global leaders, adapt successful practices to their own cultural and operational contexts, and gain a broader perspective on industry trends and best practices.
7. **Process Benchmarking:** Process benchmarking focuses on specific processes within an organisation. It involves analyzing and comparing the steps, inputs, outputs, and performance metrics of a particular process with those of leading organisations. The aim is to identify process inefficiencies, bottlenecks, and opportunities for improvement.

Failure Mode and Effects Analysis (FMEA)

Introduction to Failure Mode and Effects Analysis (FMEA)

There are numerous high-profile examples of product recalls resulting from poorly designed products and/or processes. These failures are debated in the public forum with manufacturers, service providers and suppliers being depicted as incapable of providing a safe product. Failure Mode and Effects Analysis, or FMEA, is a methodology aimed at allowing organizations to anticipate failure during the design stage by identifying all of the possible failures in a design or manufacturing process.

Developed in the 1950s, FMEA was one of the earliest structured reliability improvement methods. Today it is still a highly effective method of lowering the possibility of failure.

What is Failure Mode and Effects Analysis (FMEA)

Failure Mode and Effects Analysis (FMEA) is a structured approach to discovering potential failures that may exist within the design of a product or process.

Failure modes are the ways in which a process can fail. Effects are the ways that these failures can lead to waste, defects or harmful outcomes for the customer. Failure Mode and Effects Analysis is designed to identify, prioritize and limit these failure modes.

FMEA is not a substitute for good engineering. Rather, it enhances good engineering by applying the knowledge and experience of a Cross Functional Team (CFT) to review the design progress of a product or process by assessing its risk of failure.

There are two broad categories of FMEA, **Design FMEA (DFMEA)** and **Process FMEA (PFMEA)**.

Design FMEA

Design FMEA (DFMEA) explores the possibility of product malfunctions, reduced product life, and safety and regulatory concerns derived from:

- Material Properties
- Geometry
- Tolerances
- Interfaces with other components and/or systems
- Engineering Noise: environments, user profile, degradation, systems interactions

Process FMEA

Process FMEA (PFMEA) discovers failure that impacts product quality, reduced reliability of the process, customer dissatisfaction, and safety or environmental hazards derived from:

- Human Factors
- Methods followed while processing
- Materials used
- Machines utilized

- Measurement systems impact on acceptance
- Environment Factors on process performance

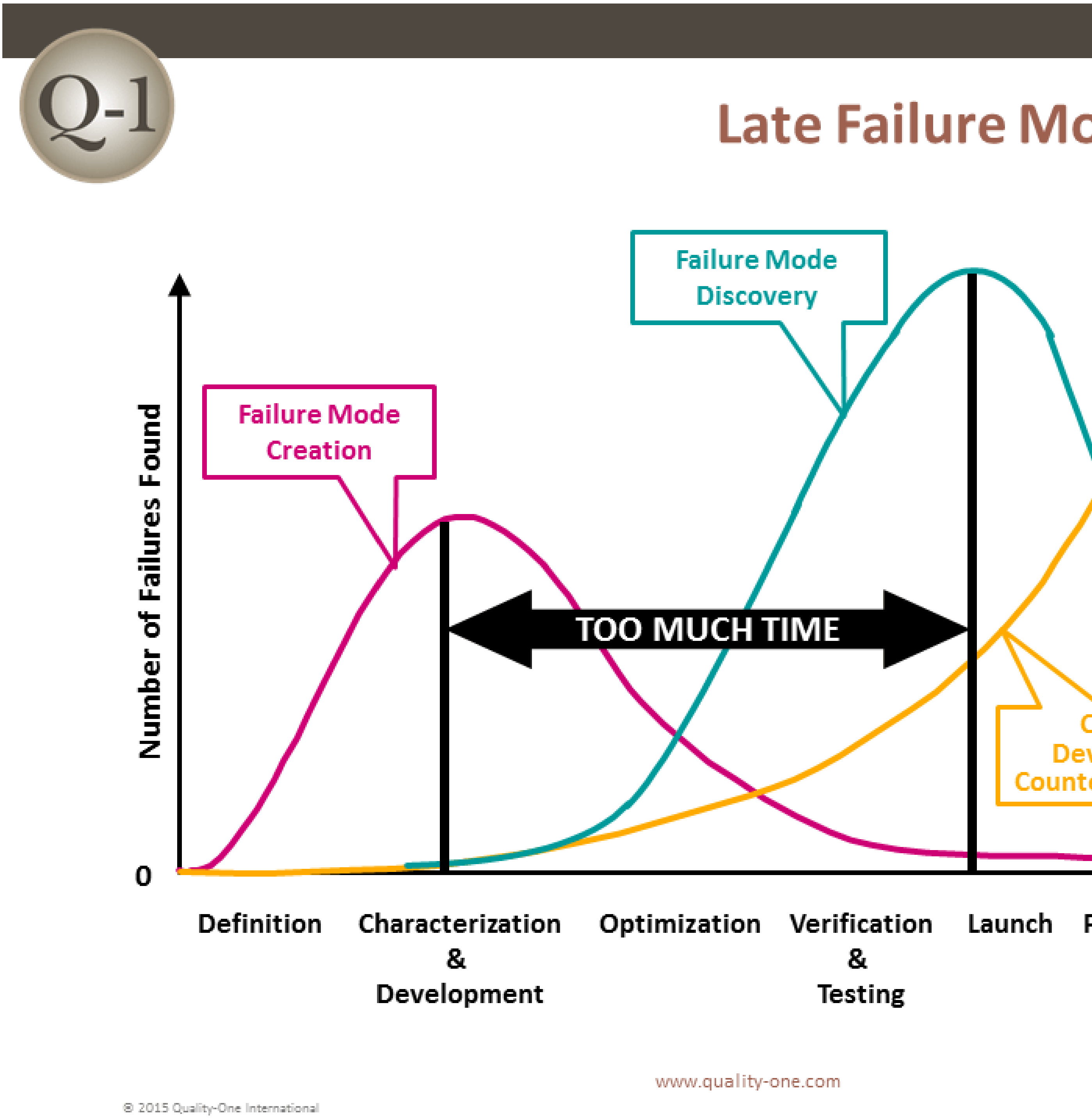
Why Perform Failure Mode and Effects Analysis (FMEA)

Historically, the sooner a failure is discovered, the less it will cost. If a failure is discovered late in product development or launch, the impact is exponentially more devastating.

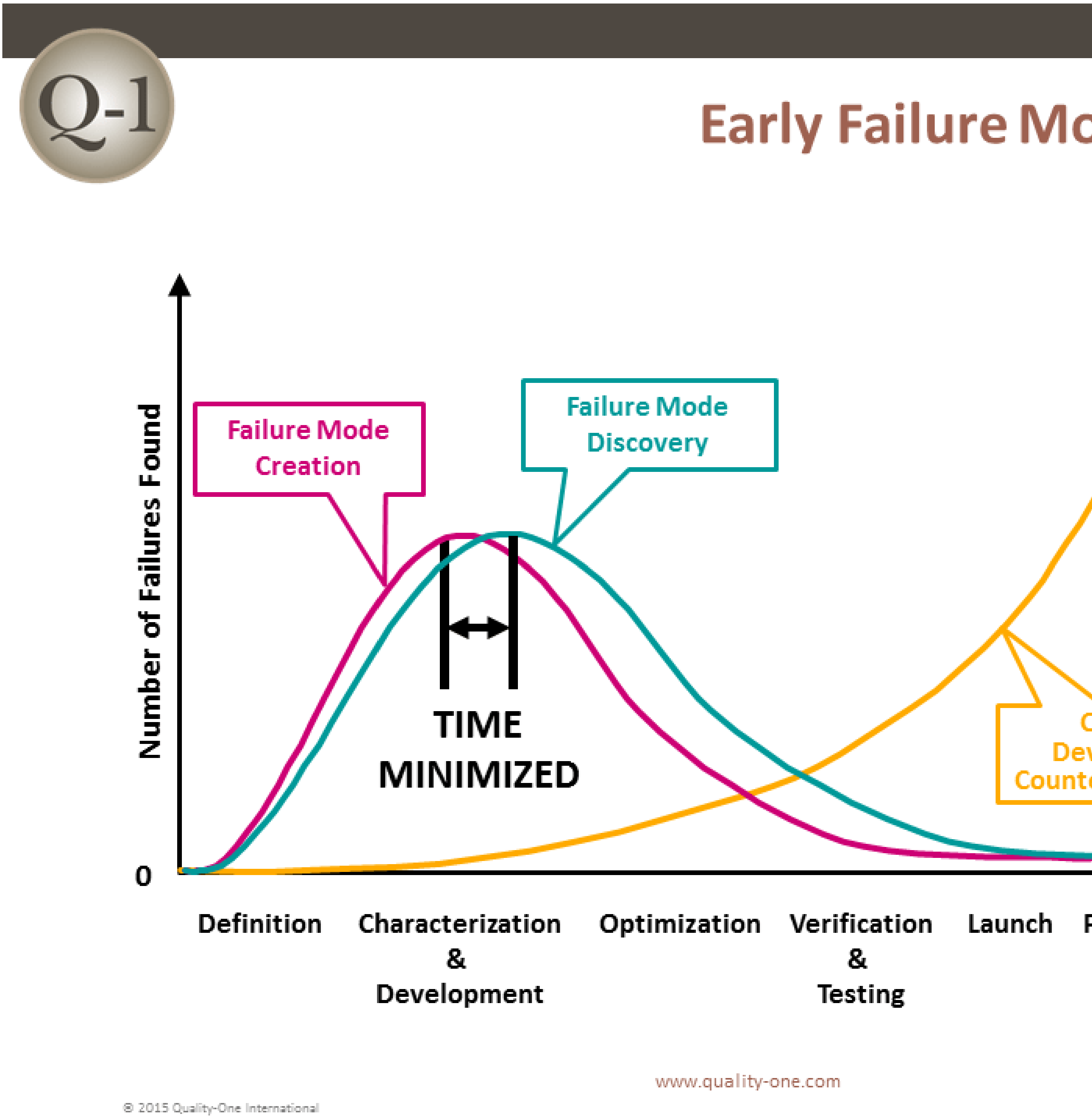
FMEA is one of many tools used to discover failure at its earliest possible point in product or process design. Discovering a failure early in Product Development (PD) using FMEA provides the benefits of:

- Multiple choices for Mitigating the Risk
- Higher capability of Verification and Validation of changes
- Collaboration between design of the product and process
- Improved **Design for Manufacturing and Assembly (DFM/A)**
- Lower cost solutions
- Legacy, Tribal Knowledge, and Standard Work utilization

Ultimately, this methodology is effective at identifying and correcting process failures early on so that you can avoid the nasty consequences of poor performance.



Late Failure Mode Discovery



Early Failure Mode Discovery

When to Perform Failure Mode and Effects Analysis (FMEA)

There are several times at which it makes sense to perform a Failure Mode and Effects Analysis:

- When you are designing a new product, process or service
- When you are planning on performing an existing process in a different way
- When you have a quality improvement goal for a specific process
- When you need to understand and improve the failures of a process

In addition, it is advisable to perform an FMEA occasionally throughout the lifetime of a process. Quality and reliability must be consistently examined and improved for optimal results.

How to Perform Failure Mode and Effects Analysis (FMEA)

FMEA is performed in seven steps, with key activities at each step. The steps are separated to assure that only the appropriate team members for each step are required to be present. The FMEA approach used by Quality-One has been developed to avoid typical pitfalls which make the analysis slow and ineffective. The Quality-One Three Path Model allows for prioritization of activity and efficient use of team time.

There are Seven Steps to Developing an FMEA:

1. FMEA Pre-Work and Assemble the FMEA Team
2. Path 1 Development (Requirements through Severity Ranking)
3. Path 2 Development (Potential Causes and Prevention Controls through Occurrence Ranking)
4. Path 3 Development (Testing and Detection Controls through Detection Ranking)
5. Action Priority & Assignment
6. Actions Taken / Design Review
7. Re-ranking RPN & Closure

The Steps for conducting FMEA are as follows:

1. FMEA Pre-Work and Assembly of the FMEA Team

Pre-work involves the collection and creation of key documents. FMEA works smoothly through the development phases when an investigation of past failures and preparatory documents is performed from its onset. Preparatory documents may include:

- Failure Mode Avoidance (FMA) Past Failure
  - **Eight Disciplines of Problem Solving (8D)**
- Boundary/Block Diagram (For the DFMEA)
- Parameter Diagram (For the DFMEA)
- Process Flow Diagram (For the PFMEA)
- Characteristics Matrix (For the PFMEA)

A pre-work Checklist is recommended for an efficient FMEA event. Checklist items may include:

- Requirements to be included
- Design and / or Process Assumptions
- Preliminary Bill of Material / Components
- Known causes from surrogate products
- Potential causes from interfaces
- Potential causes from design choices
- Potential causes from noises and environments
- Family or Baseline FMEA (Historical FMEA)
- Past Test and Control Methods used on similar products

2. Path 1 Development- (Requirements through Severity Ranking)

Path 1 consists of inserting the functions, failure modes, effects of failure and Severity rankings. The pre-work documents assist in this task by taking information previously captured to populate the first few columns (depending on the worksheet selected) of the FMEA.

- Functions should be written in verb-noun context. Each function must have an associated measurable. Functions may include:
  - Wants, needs and desires translated
  - Specifications of a design
  - Government regulations
  - Program-specific requirements
  - Characteristics of product to be analyzed
  - Desired process outputs
- Failure Modes are written as anti-functions or anti-requirements in five potential ways:
  - Full function failure
  - Partial / degraded function failure
  - Intermittent function failure
  - Over function failure
  - Unintended function failure
- Effects are the results of failure, where each individual effect is given a Severity ranking. Actions are considered at this stage if the Severity is 9 or 10
  - Recommended Actions may be considered that impact the product or process design addressing Failure Modes on High Severity Rankings (Safety and Regulatory)

3. Path 2 Development – (Potential Causes and Prevention Controls through Occurrence Ranking)

   Causes are selected from the design / process inputs or past failures and placed in the Cause column when applicable to a specific failure mode. The columns completed in Path 2 are:

- Potential Causes / Mechanisms of Failure
- Current Prevention Controls (i.e. standard work, previously successful designs, etc.)
- Occurrence Rankings for each cause
- Classification of Special Characteristics, if indicated
- Actions are developed to address high risk Severity and Occurrence combinations, defined in the Quality-One Criticality Matrix

4. Path 3 Development- (Testing and Detection Controls through Detection Ranking)

   Path 3 Development involves the addition of Detection Controls that verify that the design meets requirements (for Design FMEA) or cause and/or failure mode, if undetected, may reach a customer (for Process FMEA).

- The columns completed in Path 3 are:
  - Detection Controls
  - Detection Ranking
- Actions are determined to improve the controls if they are insufficient to the Risks determined in Paths 1 and 2. Recommended Actions should address weakness in the testing and/or control strategy.
- Review and updates of the **Design Verification Plan and Report (DVP&R)** or **Control Plans** are also possible outcomes of Path 3.

5. Action Priority & Assignment

   The Actions that were previously determined in Paths 1, 2 or 3 are assigned a Risk Priority Number (RPN) for action follow-up.

   RPN is calculated by multiplying the Severity, Occurrence and Detection Rankings for each potential failure / effect, cause and control combination. Actions should not be determined based on an RPN threshold value. This is done commonly and is a practice that leads to poor team behavior. The columns completed are:

- Review Recommended Actions and assign RPN for additional follow-up
- Assign Actions to appropriate personnel
- Assign action due dates

6. Actions Taken / Design Review

   FMEA Actions are closed when counter measures have been taken and are successful at reducing risk. The purpose of an FMEA is to discover and mitigate risk. FMEAs which do not find risk are considered to be weak and non-value added. Effort of the team did not produce improvement and therefore time was wasted in the analysis.

7. Re-Ranking RPN and Closure

   After successful confirmation of Risk Mitigation Actions, the Core Team or Team Leader will re-rank the appropriate ranking value (Severity, Occurrence or Detection). The new rankings will be multiplied to attain the new RPN. The original RPN is compared to the revised RPN and the relative improvement to the design or process has been confirmed. Columns completed in Step 7:

- Re-ranked Severity
- Re-ranked Occurrence
- Re-ranked Detection
- Re-ranked RPN
- Generate new Actions, repeating Step 5, until risk has been mitigated
- Comparison of initial RPN and revised RPN

FMEA Document Analysis

Deciding when to take an action on the FMEA has historically been determined by RPN thresholds. Quality-One does not recommend the use of RPN thresholds for setting action targets. Such targets are believed to negatively change team behavior because teams select the lowest numbers to get below the threshold and not actual risk, requiring mitigation.

The analysis of an FMEA should include multiple level considerations, including:

- Severity of 9 / 10 or Safety and Regulatory alone (Failure Mode Actions)
- Criticality combinations for Severity and Occurrence (Cause Actions)
- Detection Controls (Test and Control Plan Actions)
- RPN Pareto

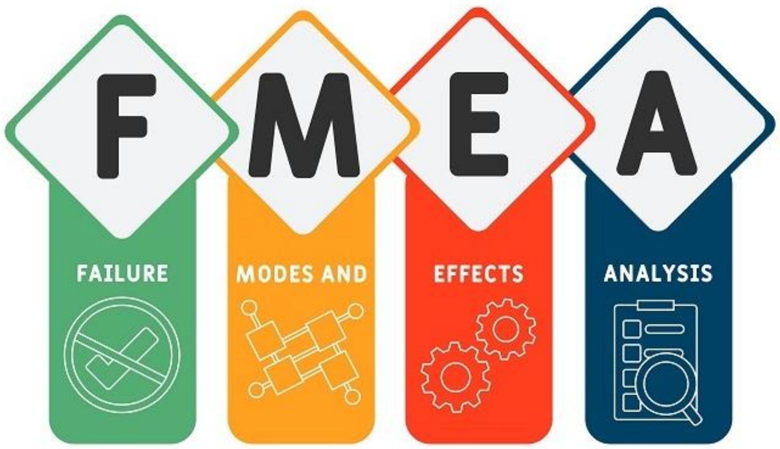
When completed, Actions move the risk from its current position in the Quality-One FMEA Criticality Matrix to a lower risk position.

FMEA Relationship to Problem Solving

The Failure Modes in a FMEA are equivalent to the Problem Statement or Problem Description in **Problem Solving**. Causes in a FMEA are equivalent to potential root causes in Problem Solving. Effects of failure in a FMEA are Problem Symptoms in Problem Solving. More examples of this relationship are:

- The problem statements and descriptions are linked between both documents. Problem solving methods are completed faster by utilizing easy to locate, pre-brainstormed information from an FMEA.
- Possible causes in an FMEA are immediately used to jump start Fishbone or Ishikawa diagrams. Brainstorming information that is already known is not a good use of time or resources.

- Data collected from problem solving is placed into an FMEA for future planning of new products or process quality. This allows an FMEA to consider actual failures, categorized as failure modes and causes, making the FMEA more effective and complete.
- The design or process controls in an FMEA are used in verifying the root cause and Permanent Corrective Action (PCA).
- The FMEA and Problem Solving reconcile each failure and cause by cross documenting failure modes, problem statements and possible causes.



Failure Mode and Effects Analysis (FMEA) is a tool used to identify potential risks associated with a process, product, or service. It is used to improve the quality of products and services by proactively reducing the risk of failure. FMEA involves analyzing each step in a process for potential causes that could lead to undesirable outcomes.

This analysis helps organizations prioritize possible causes based on the severity of the impact they would have if an issue were to occur. Each step in the process is then rated according to its likelihood of failure and how serious it would be if it happened.

The goal is to identify high-risk areas so that appropriate measures can be taken before problems arise. By analyzing processes ahead of time, companies can save money and resources in addition to increasing customer satisfaction by providing higher-quality products or services.

Types of FMEA

Design FMEA (DFMEA) is a technique used to identify potential design-related failures and their effects. This type of FMEA focuses on the product design, or engineering aspects such as materials, components, processes, and features. It helps organizations reduce errors in the early stages of development by analyzing possible problems before they occur.
Process FMEA (PFMEA) is an analytical tool used to assess the risks associated with each step within a process flow. It identifies potential failure modes based on how well the process has been designed and implemented. By analyzing all steps in a process, PFMEAs can help organizations anticipate what could go wrong and prevent costly mistakes from occurring downstream. In addition to providing risk assessment for existing processes, it can also be used to improve new designs before manufacture begins.

Steps in FMEA Process

Establish the FMEA Team

The team should include a cross-functional, multi-disciplinary group of individuals with expertise in the product/process being analyzed. They are responsible for identifying and analyzing potential failures, their causes and effects, as well as developing corrective actions to reduce risk. This step is important because it ensures that all necessary perspectives are included in the FMEA process for a more thorough analysis.

Define the Process

This step involves defining the process to be analyzed in terms of its scope and objectives so that the team can focus their efforts.

Identify Failure Modes

This involves brainstorming to identify potential failure modes for each component of the process or product, with input from design engineers and other experts. For each identified failure mode, consider its effects on the customer's requirements as well as its likelihood of occurrence.
It is important to document all these factors in an FMEA Worksheet so that they can be tracked and easily modified over time if necessary.

Determine the Effects of Failure

This includes identifying the potential failure mode, how it could occur, and what effects it may have on customers or end users. It is important to consider every possible effect of a failure as this helps in developing effective corrective action plans.
Additionally, each potential failure should be ranked according to its severity and criticality in order to prioritize corrective actions that are necessary for a successful product design or process improvement activity.
Failure Mode Analysis also looks at the root cause of any identified problem so that future occurrences can be prevented or minimized by implementing preventive measures early on in the development process. Finally, an analysis of the potential risks associated with any proposed solution must also be conducted before determining if it is worth pursuing further.

Identify Causes and Prevention Measures

This can be done through brainstorming sessions with experts or looking at similar products for design ideas. During this step, it is important to look at all aspects of the system including hardware, software, processes, and user inputs. After identifying the potential causes of failure, prevention measures should be identified so that any risks associated with these failures can be mitigated or eliminated altogether.
The next step in the FMEA process is to evaluate each risk based on its severity, probability of occurrence, and ease of detection. Each risk will then receive an RPN score which indicates how serious it is relative to other risks within the system.

Once all risks have been evaluated, they should be prioritized according to their RPN scores so that corrective actions can begin as soon as possible on those items deemed most critical for preventing failure modes from occurring again in future iterations or versions of your product or service offering.

Update the FMEA

Updating the FMEA is a process of reviewing and revising the previous version, based on new findings or data collected. It includes changing RPN (Risk Priority Number) values if necessary, adding more corrective actions to bring down any potential failure modes or effects that are discovered since the last review, and making other changes as needed.
After updating an FMEA document, it should be signed off by stakeholders before it can be considered complete. This ensures that all parties agree with the results of the analysis and any recommended actions for reducing risk in future designs or processes.

Benefits of FMEA

Failure Mode and Effects Analysis (FMEA) is a powerful tool used to identify potential failure areas in processes, products, services, or software. It helps organizations uncover problems before they become actual failures, allowing them to address issues proactively instead of reactively.
FMEA also helps organizations improve their design process by identifying opportunities for improvements. Benefits of implementing FMEA include improved reliability and safety with fewer customer complaints, reduced costs associated with warranty claims and repairs, improved product quality through better defect prevention strategies, increased customer satisfaction due to more reliable products/services/software, and a reduced risk of liability from faulty products or services.
Additionally, it can help organizations meet regulatory standards since the analysis highlights any non-conformities that could result in fines or other penalties.
Overall, FMEA provides invaluable insights into how an organization can improve its design process while reducing risks associated with defective products or services.

Conclusion

All in all, Failure Mode and Effects Analysis (FMEA) is a valuable tool for any organization to use. Not only does it help to identify potential problems before they can cause harm or disruption, but it also helps organizations develop more effective processes and procedures that will mitigate the risks associated with those problems.
By using FMEA regularly, companies can ensure their operations are running as efficiently and effectively as possible while minimizing risk. Additionally, FMEA helps an organization build better relationships with customers by demonstrating a proactive approach toward preventing errors and improving product quality.
With these advantages in mind, adopting a Failure Mode and Effects Analysis strategy should be considered essential for any business seeking long-term success.

What is QFD?

Quality Function Deployment (QFD) is a proactive, focused approach used in taking the Voice of the Customer (VOC) into account and then carrying out effective responses to customer expectations and needs during the design and production stages. A key component of QFD is the House of Quality (HoQ) matrix, which helps translate customer requirements into specific engineering or operational goals. Initially considered a form of cause-and-effect analysis, QFD is an essential tool not just for quality management but also for planning.

Why Implement QFD in Your Organization

The goal was to create a methodology that integrates customer satisfaction into a product before implementing its manufacturing process. This presents a more proactive approach than other methods of quality control that focus on fixing customer issues during or after manufacturing.

Usually, customer requirements are communicated in the form of qualitative terms such as "safe," "user friendly," "ergonomic," and so on. For organizations to effectively develop a product or service, these qualitative terms from customers need to be translated into quantitative design requirements. This is where QFD comes in. By combining various matrices to form a comprehensive house-like structure called the "House of Quality", manufacturers and production facilities can come up with a prioritized list of customer needs and pain points and a systematic process of integrating them into the design of the product or service.

Hence, organizations can then make better decisions after weighing customer requirements and considering how the products or services go against the competition in the market. How QFD helps in this aspect is that it provides the needed documentation for the overall decision-making process.

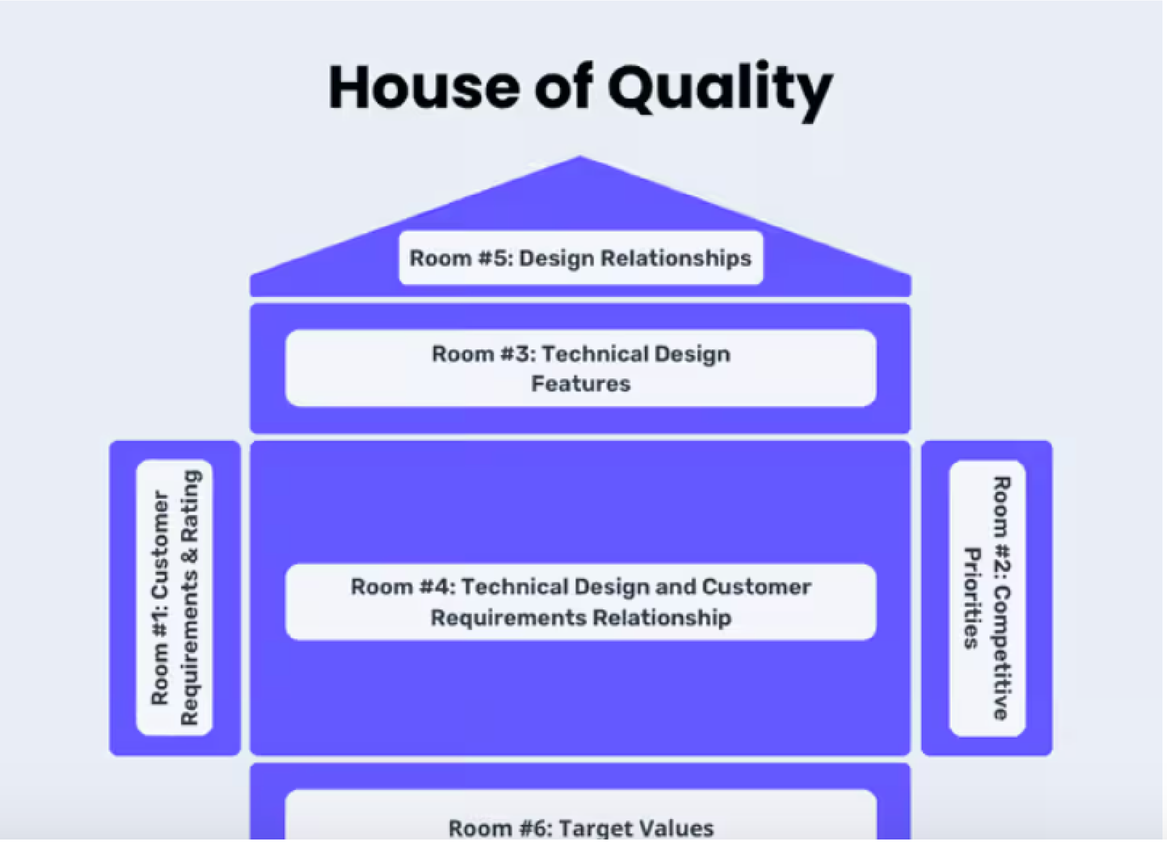
Benefits of QFD

Now, how does this approach help organizations ensure the quality of their processes, particularly in product development? To summarize, here are a few of the major advantages of using QFD:

- **Gain a better understanding of your customers** – Be able to predict, know, and act on what they need, either stated or unstated, and provide a better customer experience.
- **Utilize customer feedback for continuous improvement** – Transform what the customers need into quantifiable business goals through the form of performance metrics, process controls, and more.
- **Establish a better structure of requirements** – The various stages of product development require efforts across departments, and inefficiencies in the process may occur. QFD helps build a standardized system that minimizes unnecessary risks and bottlenecks.
- **Have an efficient way of allocating resources** – By knowing the priority list of the customer requirements to be implemented, organizations can then be more intentional and focus on critical improvement opportunities.

What is a House of Quality Matrix?

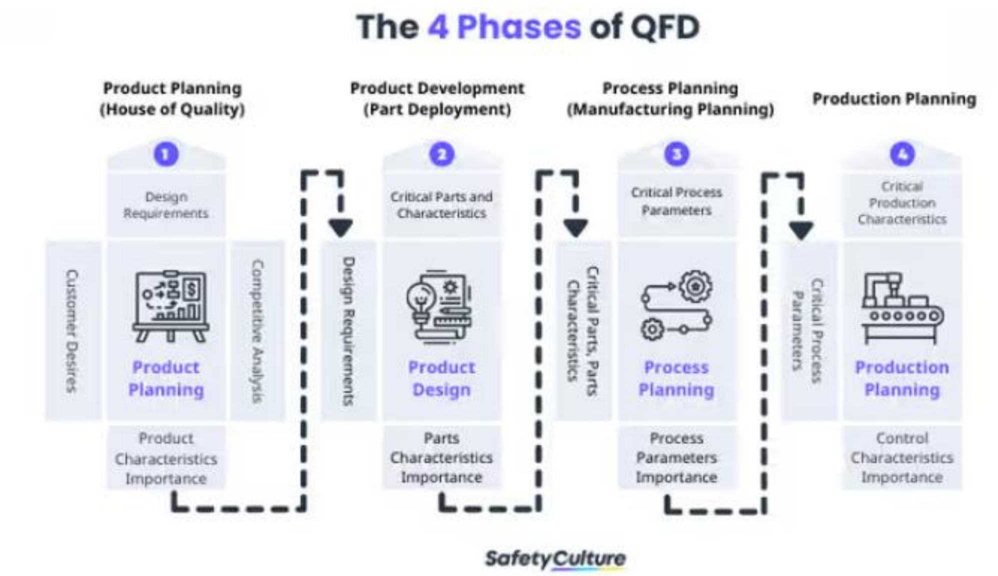
Also known as the product planning matrix, the House of Quality (HoQ) is practically the most essential stage of the QFD. This is where every bit of detail must be taken into consideration, as it's going to set the pace for the succeeding stages of the four-phase QFD. Further, HoQ serves as a primary graphical representation of the product planning and development process.



What are the 4 Phases of QFD?

QFD enables cross-functional teams to collaborate and find a balance between customer needs and technical requirements. This is crucial in streamlining the way manufacturers and production facilities decide what kind of features or characteristics they should include in their products and services that customers will buy and value.

The comprehensive process of QFD is divided into 4 major phases: Product Planning, Product Development, Process Planning, and Production Planning. Each matrix or visual representation used under each of the 4 phases is related to the previous one. Let's look into how to implement QFD into your business process and what happens at each stage.



The 4 Phases of QFD

1. Product Planning

In this stage, the House of Quality matrix is used to translate customer needs into design requirements. The Voice of the Customer (VOC) is gathered through conducting Focus Group Discussions (FGDs), interviews, and other similar methods using Voice of the Customer templates. Then, feedback and insights from such customer requirements are lined up for design and product features considerations.

As this phase will help set the pacing of the entire QFD process, it's important to cover various aspects as well. These include identifying the underlying—apart from the obvious, stated ones—customer requirements, characteristics of possible market offerings, and competitive opportunities. It’s also crucial to note that this stage is typically led by the Marketing department.

2. Product Development

This phase enables the team to specify design requirements by identifying critical parts and assembly components. These are then based on the prioritized list of offering characteristics gathered in the Product Planning phase using the House of Quality matrix.

3. Process Planning

The third phase involves deciding on the design manufacturing and assembly process that best addresses the requirements discussed in the Product Development phase. Teams responsible for this aspect must be able to define the required operational guidelines, elements, and parameters.

4. Production Planning

The final phase is where process control methods are established, along with the production and inspection processes. These are key to ensuring that the characteristics are met and continuously undergo evaluation and improvement.

Quality Function Deployment: The Customer-Driven Methodology

Customer happiness is the key for any business that wants to succeed and to stay in the long run. One approach that helps companies listen to and please their customers is called Quality Function Deployment, or QFD for short.

QFD gives teams a powerful way to understand exactly what clients want and need when designing products or processes.

It works by translating vague wants into clear objectives right from the beginning. Quality Function Deployment uses special charts and tables to change vague ideas into actual steps that can be measured.

This customer-focused process began in the 1960s in Japan during a big push for higher quality. Companies like Toyota were among the first to use it, and soon other industries saw benefits too by using QFD to create exactly what customers were seeking.

While it started in manufacturing, QFD quickly spread to all sorts of businesses as an easy way to gain an edge through more satisfied repeat customers.

Key Highlights

- Understand the fundamentals of Quality Function Deployment (QFD) and its philosophy
- Explore the rich history and global adoption of QFD as a catalyst for innovation
- Learn the key benefits of Quality Function Deployment
- Master the four phases of Quality Function Deployment: product planning, product design, process design, and quality control
- Dive deep into essential QFD tools
- Gain insights on applications of QFD across diverse industries like automotive, manufacturing, technology, and services
- Learn best practices for implementing Quality Function Deployment in your organization
- Discover how to overcome common challenges and pitfalls when deploying QFD

What is Quality Function Deployment (QFD)?

In Quality Function Deployment lies a fundamental question – how can organizations truly understand and translate the voice of the customer into tangible product and process designs that consistently exceed expectations?

As a disciplined methodology, QFD provides a robust framework for capturing customer requirements, prioritizing them based on importance, and systematically deploying them throughout the entirety of the development lifecycle.

For teams looking to strengthen their quality management toolkit, combining QFD with a Six Sigma certification can provide a powerful synergy—leveraging QFD's customer-centric planning alongside Six Sigma's data-driven process improvement techniques.

Voice of the Customer (VOC)

The first step in any QFD initiative is to thoroughly research and document the voice of the customer.

This involves gathering qualitative and quantitative insights into what customers desire, need, and value in a product or service offering.

Common techniques include surveys, interviews, focus groups, analyzing sales data and customer feedback, ethnographic studies, and more.

The goal is to transform these voices into specific, actionable customer requirements or "**whats**" that can drive the design process.

Customer Requirements

Once the voice of the customer data has been collected and analyzed, the next step is to distill it into a prioritized list of customer requirements.

These requirements articulate the features, functions, performance metrics, and non-functional attributes that customers deem essential or desirable.

House of Quality

The House of Quality matrix is arguably the most powerful and recognizable tool in the QFD toolbox.

This complex diagram maps customer requirements against technical specifications, design characteristics, competitive benchmarks, and more.

Matrix Product Planning

Quality Function Deployment is a matrix-driven planning methodology that cascades requirements and priorities across multiple dimensions of the product realization process.

Subsequent matrices then deploy these outputs into increasing levels of technical detail for product design, manufacturing process design, production planning, and quality assurance.

Customer-driven Engineering

The overarching philosophy of QFD is to embed the voice of the customer into every engineering decision, from early product conceptualization through full-scale production.

Rather than relying on assumptions or historically based requirements, customer-driven engineering ensures that limited resources are directed toward delivering maximum customer value and satisfaction.

History and Origins of Quality Function Deployment (QFD)

Japanese Roots

While Quality Function Deployment has since been embraced globally, its origins trace back to the Japanese quality revolution of the 1960s.

During this pivotal era, pioneers like Dr. Shigeru Mizuno and Dr. Yoji Akao recognized the need for more robust quality practices that started by deeply understanding customer desires.

Their insights laid the theoretical foundations for QFD at a time when Japanese manufacturing was disrupting industries through a laser focus on quality and customer satisfaction.

Yoji Akao and Shigeru Mizuno

Dr. Yoji Akao, a respected quality management consultant, is widely credited as the founder of Quality Function Deployment (QFD).

While assisting the shipbuilding division of Mitsubishi Heavy Industries, Akao developed the foundational matrices and prioritization processes that became QFD.

His colleague Dr. Shigeru Mizuno helped refine and document the methodology based on their experiences working with cross-functional teams in manufacturing environments.

Mitsubishi Shipyard

At Mitsubishi's Kobe shipyard, QFD was initially conceived and field-tested in the late 1960s.

Facing intense global competition, the shipbuilders desperately needed a way to systematically capture and address customer needs while making intelligent trade-offs during design and construction.

The early successful adoption of Quality Function Deployment at this shipyard demonstrated its potential for bridging the gap between customer expectations and engineering execution.

Toyota and Suppliers

While QFD's origins were in shipbuilding, it was the automotive industry that rapidly embraced and propagated the methodology across Japan in the 1970s and 80s.

The Toyota Motor Company, already renowned for its Quality Circles and just-in-time production principles, became an early advocate of QFD.

Toyota mandated that its vast supplier network also adopt Quality Function Deployment, catalyzing its widespread adoption as a cornerstone of Japanese quality practices.

US Adoption in the 1980s

As Japanese automakers gained significant market share in the US during the 1970s, the Detroit Big Three desperately sought a competitive response.

QFD was identified as a key enabler of the quality renaissance that allowed Japanese manufacturers to meet and exceed customer expectations so consistently.

By the early 1980s, Ford, GM, and Chrysler had all sent teams to study Quality Function Deployment practices in Japan. Their subsequent adoption and customization of QFD for American markets sparked its proliferation across other industries.

The Importance and Benefits of Quality Function Deployment

In my decades of experience deploying Quality Function Deployment across diverse organizations, QFD represents a fundamental shift in mindset – from operating based on internal assumptions and historical requirements, to proactively capturing and responding to the voice of the customer.

When properly implemented, the benefits of QFD are both quantifiable and wide-ranging, impacting everything from customer satisfaction and loyalty to development cost and time-to-market.

Customer Satisfaction

Perhaps the most obvious yet critical benefit of QFD is its ability to improve customer satisfaction levels dramatically.

By systematically deploying specific customer requirements throughout the design process, organizations ensure that products and services more precisely align with what users want and value.

Design Quality

In addition to satisfied customers, Quality Function Deployment also drives higher internal quality levels by reducing defects and rework.

When customer requirements are documented and flowed down to granular engineering specs, there is far less ambiguity and need for assumptions during design and manufacturing, a principle that professionals with six sigma certification often apply using tools to minimize variation and defects.

Competitive Analysis

A core tenet of QFD is performing rigorous competitive benchmarking during the customer requirement capture phase.

By evaluating how effectively competing products and services address customer needs, organizations can pinpoint whitespace opportunities.

The voice of the customer data highlights where incumbents are underdelivering, allowing new offerings to be precisely targeted at those gaps and rapidly gain market share through superior quality.

Key Advantages of Quality Function Deployment

Customer-focused Products

QFD is a customer-focused product development and quality methodology.

By making the customer requirements and prioritized needs the driving input, QFD ensures that limited resources are channeled into delivering maximum customer-perceived value.

Shorter Development Cycles

Another major advantage of Quality Function Deployment implementations that I've witnessed is significantly accelerated product development timelines.

Organizations often pair QFD with a Six Sigma Green Belt certification to further streamline workflows, ensuring both customer needs and process efficiency are prioritized.

Because customer requirements are documented upfront and flow through the design process, there is far less ambiguity and need for rework.

Cross-functional Collaboration

Effective QFD requires input and buy-in from cross-functional teams spanning marketing, engineering, operations, quality, and more.

The process of capturing the voice of the customer, prioritizing needs, and mapping them to technical requirements is inherently collaborative.

QFD breaks down organizational silos and fosters greater cross-pollination of knowledge – a critical enabler for developing successful products that delight customers.

Structured Documentation

A hallmark of QFD compared to more ad-hoc product development approaches is its structured and documented nature.

Each phase leaves behind a detailed paper trail outlining the key inputs, analysis, trade-offs, and prioritization decisions made.

This documentation creates an invaluable organizational knowledge repository that can be leveraged for future programs and continuous improvement initiatives.

Prioritized Requirements

In the real world of finite resources and complex development constraints, not all customer requirements can be fully addressed.

This is where Quality Function Deployment's ability to rigorously prioritize requirements based on importance and customer preferences becomes so powerful.

By focusing delivery efforts on the highest impact requirements first, organizations can maximize customer satisfaction while balancing development risks and costs.

The Four Phases of Quality Function Deployment

While the core principles of QFD are powerful, the true strength of this methodology lies in its structured, phased approach to embedding the voice of the customer throughout the entire product realization process.

The QFD workflow is commonly divided into four distinct phases, each building upon the previous one to translate high-level customer needs into increasingly granular engineering requirements, manufacturing processes, and quality controls.

Phase 1 – Product Planning

Voice of the Customer (VOC) research

The product planning phase begins with a concentrated effort to capture the voice of the customer through qualitative and quantitative research techniques like surveys, interviews, focus groups, and analyzing past complaints and desires.

This step is critical for unearthing both obvious requirements that customers articulate, as well as latent, unstated needs.

Customer Needs and Wants

The raw VOC data is then synthesized and distilled into a hierarchical list of customer needs, wants, and preferences ranked by their relative importance.

Effective need statements are precise, measurable, and accurately convey the intent behind the customer's desired outcome or performance expectation.

Technical Specifications

With prioritized customer needs in hand, cross-functional teams can begin mapping them to specific technical specifications, features, and design characteristics that the product must deliver.

These “**hows**” form the basis for more detailed engineering requirements in subsequent phases.

Relationship matrix

The iconic House of Quality matrix visualizes the relationships between each customer’s need and the proposed technical specifications.

This allows teams to identify potential conflicts, make informed trade-offs, and prioritize which specs will yield the greatest impact on customer satisfaction based on their importance scores.

Competitive Analysis

An integral part of the House of Quality is a competitive assessment section that scores how effectively competing products address each customer’s need.

This benchmarking highlights potential gaps and market opportunities to be exploited in the new design.

Phase 2 – Product Development

Critical Parts and Assemblies

With high-level technical specifications defined, the product development phase dives deeper to identify the critical product sub-systems, components, and assemblies required to deliver on those specs.

Each one is systematically mapped and traced back to its impact on specific customer needs.

Key characteristics

For each critical part or assembly, key characteristics that contribute to fulfilling technical requirements are documented.

These may include form, fit, function, performance metrics, and more. Detailed engineering targets, tolerances, and acceptance criteria are established.

Product Design Requirements

The aggregated key characteristics from all components flow down to become the product design requirements and a DFMEA is conducted to identify and mitigate potential failures in delivering on customer needs.

Design FMEA Inputs

The relationship matrices and prioritized customer needs from the previous phase feed directly into the Design Failure Mode and Effects Analysis (DFMEA) process.

This allows the most critical customer requirements to drive risk identification and mitigation strategies during design.

Phase 3 – Process Planning

Manufacturing Processes

With the product design defined, teams can now determine the specific manufacturing processes required to produce components and assemble the final product according to specs.

Process maps, flow diagrams, workstations, and procedures are documented.

Process Flow

Each step in the manufacturing process flow is analyzed to understand potential impacts on key product characteristics.

This evaluation highlights critical process parameters that must be carefully monitored and controlled.

Critical Process Characteristics

Through mapping, the team identifies the vital few process characteristics that contribute most significantly to producing conforming components and assemblies.

These critical inputs become the focal points for statistical process control and real-time monitoring.

Quality Controls

With critical process parameters identified, quality teams can establish rigorous control plans, sampling strategies, and acceptance criteria.

Tools like SPC, mistake-proofing, and FMEA ensure these characteristics remain stable and in statistical control.

Phase 4 – Process Quality Control

Process Capability

Before full-scale production launch, the manufacturing processes undergo thorough capability studies and statistical validation.

This data-driven approach provides quantitative evidence that processes can reliably produce conforming products within specification limits.

Inspection and Testing

Complementing the established quality controls, comprehensive product inspection, and testing protocols are defined based on critical characteristics.

These may include incoming inspections, in-process checks, final product acceptance tests, accelerated life tests, and more.

Continuous Improvement

Quality Function Deployment is an iterative methodology anchored in the principle of continuous improvement.

As production ramps up, processes are closely monitored using statistical tools, and any deviations from expected quality targets are treated as opportunities to further optimize.

Professionals leading continuous improvement initiatives often pursue a Six Sigma Green Belt certification to master tools like control charts and defect analysis, which complement QFD's feedback loops.

The Customer Feedback Loop

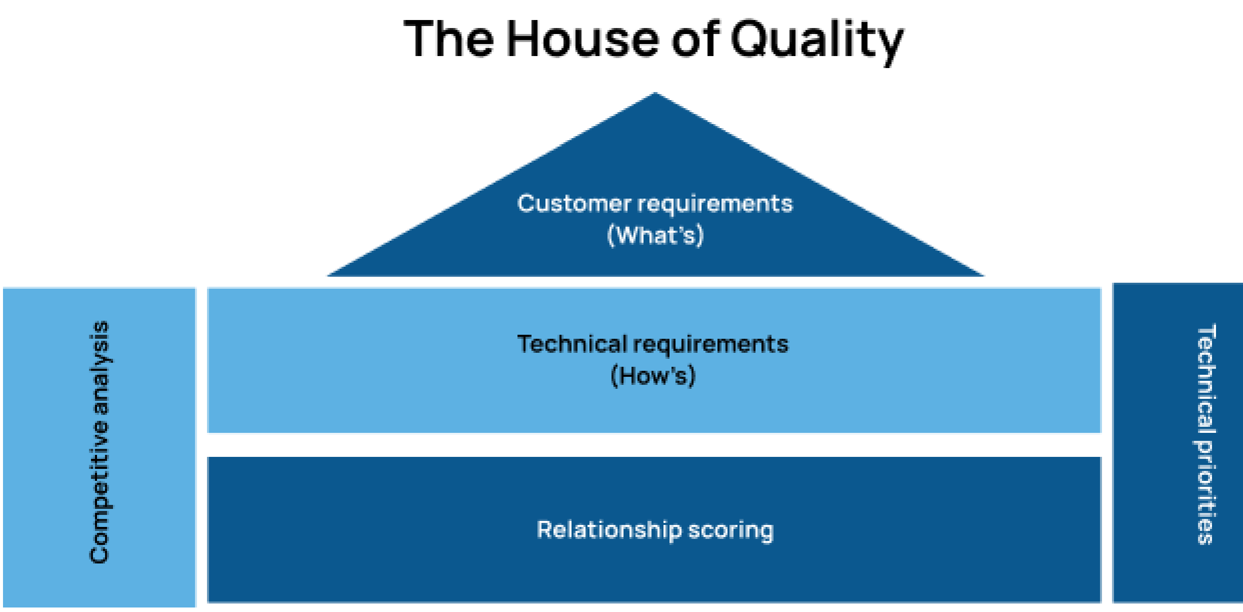
Perhaps most importantly, the QFD workflow establishes closed-loop mechanisms to constantly solicit and incorporate new customer feedback.

This voice of the customer is analyzed and prioritized, with findings deployed back through the appropriate QFD phases to drive ongoing product and process enhancements.

Quality Function Deployment Tools and Matrices

Quality Function Deployment methodology lies in a detailed set of interlinked matrices and prioritization tools.

While the House of Quality is arguably the most recognizable, the true power of QFD emanates from the structured way these matrices cascade and deploy requirements across the entire product realization cycle.



The House of Quality Matrix

Customer Requirements ("Whats")

The left side of the iconic House of Quality matrix captures the voice of the customer in the form of specific "**what**" requirements and needs.

These precise statements convey both obvious demands customers articulate as well as latent needs uncovered through research.

Critically, each "**what**" is weighted based on its relative importance to customer satisfaction and business impacts.

Technical Requirements ("Hows")

Across the top of the matrix, the team documents the key technical requirements, specifications, and design characteristics that potentially address the customer's "whats".

These "hows" define what the product or service must deliver to fulfill the associated customer needs.

Relationship Scoring

The body of the House of Quality is a relationship matrix that scores the correlation between each "what" and "how".

This criticality scoring allows the team to pinpoint the vital few technical requirements that will yield the greatest impact on achieving prioritized customer needs.

Competitive Analysis

An integral component is a competitive assessment that scores how effectively competing products and services address each "what".

This benchmarking exposes gaps and market opportunities for the new design to exploit in exceeding customer expectations.

Technical priorities

Through roll-up calculations, each "how" receives an overall importance rating based on its cumulative impact across all customer requirements.

This prioritization ensures development resources are focused on the most critical technical characteristics driving customer satisfaction.

What is the Taguchi Loss Function

One crucial and highly beneficial aspect of Six Sigma is all the statistical and analytical tools it comes with. All of them can help a business and its management gain deeper and more complete understanding of the business processes implemented, so they can be optimized and adjusted for a higher standard of results and outcomes.

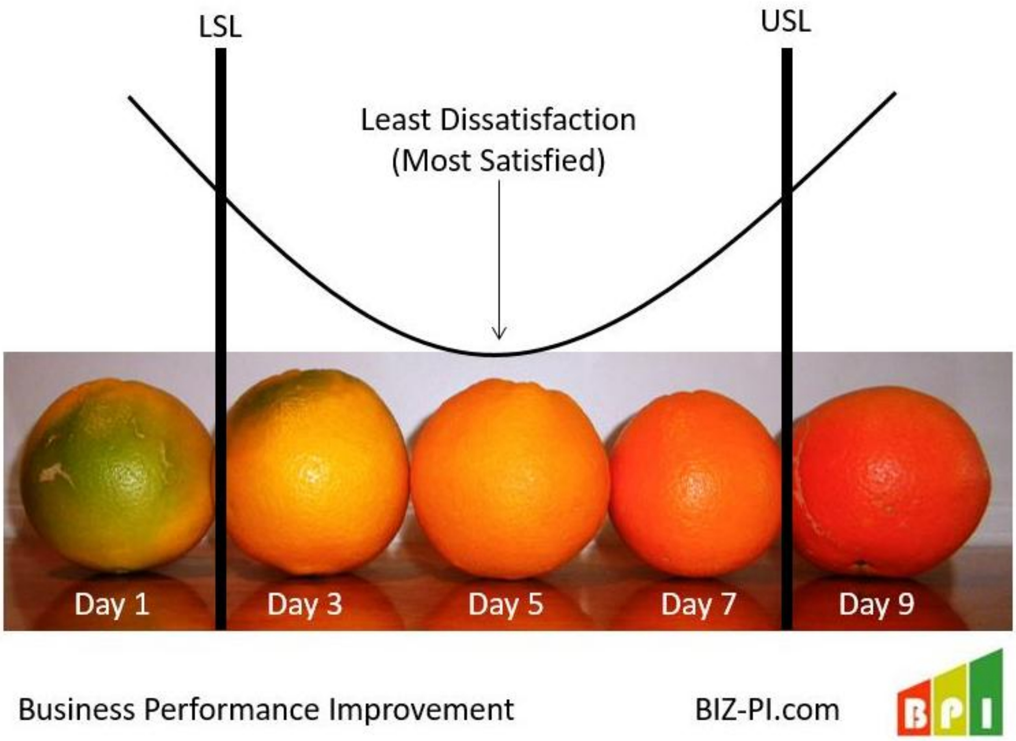
One such curious and powerful concept is the Taguchi Loss Function (named for Genichi Taguchi), which is used to assess the financial impact of deviating from the target or optimal performance level, usually set by the customer.

What does the Taguchi Loss Function measure exactly

One of the core assumptions of Six Sigma is that not allowing the outcomes of business processes to deviate from the target is bound to create value for the particular organization. As you drift away from the target, the performance of the process, product or service starts to degrade in the eyes of your customer.

When examining this concept, one can intuitively incur that there is going to be a correlation between not meeting the quality target and waste of resources. Of course, whenever resources are wasted and whenever the quality of the results from a business process is low, this is going to have some financial ramifications that can be interpreted as a loss. This is exactly what the Taguchi Loss Function measures and what it is used to represent graphically.

When the outcome of a business process is exactly as targeted with zero deviation, the Taguchi Loss Function also has a value of zero. As deviation increases, the loss incurred and measured by the function increases in a quadratic manner. Even deviation within the Six Sigma range incurs some nominal loss, and that loss does not appear suddenly at its boundary, but increases gradually with the increase in deviation.



If we use the image above, we see that the ideal orange has ripened at Day 5. This is where the customer is most satisfied (or least

dissatisfied). If we move one day earlier or later, we start to incur some dissatisfaction, as it is no longer the best orange possible. Even if it is acceptable to sell an orange on Day 3 or Day 7, those oranges are not as good as Day 5 oranges.

When is the Taguchi Loss Function useful

When a business decides to optimize a particular process, or when optimization is already in progress, it’s often easy to lose focus and strive for lowering deviation from the target as an end goal of its own. But whenever we are talking about business, the most important and the most fundamental metric is always the profit. In that sense, the Taguchi Loss Function can be extremely valuable, as it is a tool that can transform deviation from target to a value with a financial representation, which affects the bottom line directly. Going back to the orange example, the customer would be getting an orange with less value on Day 3 or Day 7, yet paying the same price as Day 5 oranges. This may lead to financial impacts of lost future sales or more customer returns.

This is why this tool is widely used when selecting business processes to optimize first. Applying the function allows for the managers and consultants to measure the financial impact that the planned process improvement would have for the organization. This means that every project can be evaluated with a specific value for its potential financial impact. which makes prioritization a much easier and more informed decision.

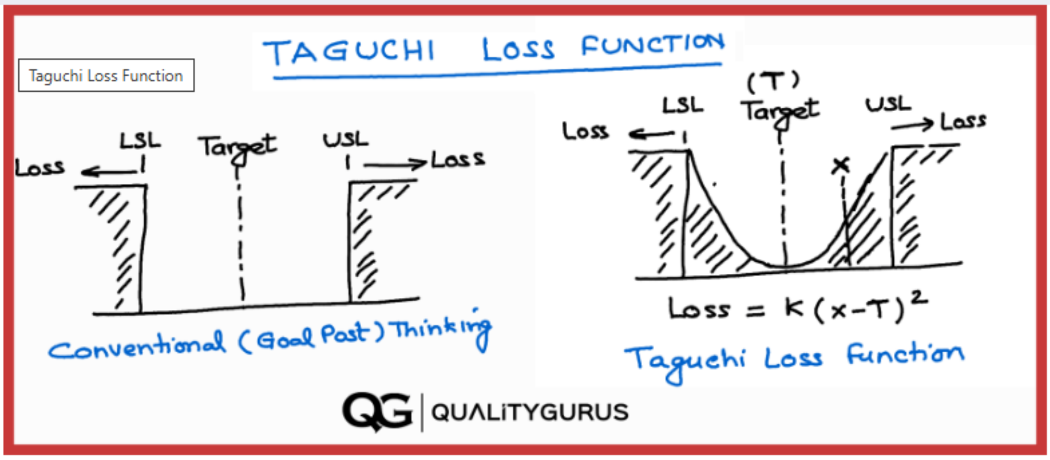
The most famous case of this concept in practice was Ford’s study of similar transmission parts manufactured by an overseas supplier (likely Mazda). They found that they had superior performing parts due to less variation, even though all parts were within specification limits.

A more general usage

Despite the fact that the Taguchi Loss Function in the context of Six Sigma is generally used to measure the correlation between process performance and financial loss, it actually doesn’t have to be limited at that. It can be used to measure and predict the resulting losses in a wide variety of other metrics like customer satisfaction, product quality and employee time utilization efficiency.

The Taguchi Loss Function is an invaluable resource when planning and strategizing. This is why it is not a tool that is utilized just by Six Sigma alone, but by a wide variety of other business process optimization methodologies like Lean and Lean Six Sigma.

What we find most valuable with Taguchi Loss Function is the concept and change in mindset that it brings. It helps people realize that variation has a financial impact, and a company should not be complacent with quality just because they are meeting the specification limits. As the video showed, reduced variation can be a competitive advantage in your industry!



In quality engineering, the **Taguchi Loss Function** holds significant importance as a key concept pioneered by Dr. Genichi Taguchi. Dr. Taguchi, a renowned Japanese engineer and statistician, was a firm believer in the idea that reducing variation in product quality is essential for achieving consistent, high-quality performance.

Dr. Taguchi's primary objective was to shift the focus from merely meeting specifications to actively minimizing variation. This approach encourages manufacturers to strive for continuous improvement and deliver consistently high-quality products that meet customer expectations. By emphasizing the minimization of variation, the Taguchi Loss Function helps drive the development of robust designs and improved manufacturing processes, ultimately leading to better customer satisfaction and reduced costs for manufacturers.

Taguchi Loss Function

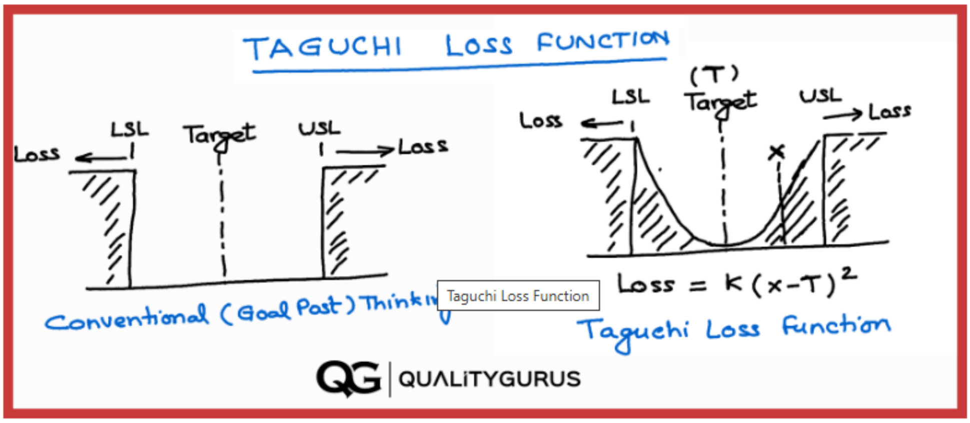
The Taguchi Loss Function (or Quality Loss Function) is a mathematical representation that quantifies the relationship between product quality and the financial loss (loss to society) associated with deviations from a target value. The Taguchi Loss Function is an equation used to quantify the financial loss incurred due to deviations in product quality from the target value. It is based on the principle that any deviation from the ideal or target value increases costs associated with poor quality, customer dissatisfaction, or product failures.

The Taguchi Loss Function equation is:

$L(x) = K * (x - T)^2$

Where:

- L(x) represents the financial loss associated with a specific deviation from the target value.
- K is the loss coefficient, a constant that determines the rate at which financial loss increases with deviations from the target value.
- x is the actual value of the quality characteristic being evaluated.
- T is the target value, the ideal or optimal value of the quality characteristic.



The equation is quadratic in nature, indicating that the financial loss increases exponentially as the deviation from the target value grows. This means that even small deviations from the target can result in significant financial losses, emphasizing the importance of maintaining high-quality products that meet customer expectations.

Example Calculation:

Suppose a company manufactures widgets, and the target weight for these widgets is 500 grams (T = 500). The company has a tolerance range of ±10 grams, meaning any widget with a weight between 490 and 510 grams is considered acceptable. Widgets outside this specification limit (acceptable limits) are rejected, costing the company $50 per rejected widget.

First, we need to calculate the loss coefficient (K) value using the provided information. We know the loss is $50 when the deviation from the target weight is 10 grams.

L(x) = K * (x - T)^2
$50 = K * (10)^2

Solving for K:

K = $50 / (10)^2
K = $50 / 100
K = $0.50

Now that we have the value of K, let's calculate the financial loss (L(x)) for a widget with a weight of 509 grams (x = 509). Please note that this piece is within the acceptable tolerance range. Conventionally we would not consider any loss (quality cost) in this case.

Using the Taguchi Loss Function equation:

L(x) = K * (x - T)^2

Using the Taguchi Loss Function equation:

L(x) = K * (x - T)^2

Plugging in the values:

L(509) = 0.50 * (509 - 500)^2
L(509) = 0.50 * (9)^2
L(509) = 0.50 * 81
L(509) = $40.50

According to the Taguchi Loss Function, the financial loss associated with producing a widget with a weight of 509 grams is $40.50. This calculation helps the company understand the cost implications of deviations from the target weight.